

only if the entry is longer than 50 characters. Use the admin site to add an entry that's fewer than 50 characters in length, and check that it doesn't have an ellipsis when viewed.

18-3. The Django API: When you write code to access the data in your project, you're writing a *query*. Skim through the documentation for querying your data at <https://docs.djangoproject.com/en/4.1/topics/db/queries>. Much of what you see will look new to you, but it will be quite useful as you start to work on your own projects.

18-4. Pizzeria: Start a new project called `pizzeria_project` with an app called `pizzas`. Define a model `Pizza` with a field called `name`, which will hold name values, such as `Hawaiian` and `Meat Lovers`. Define a model called `Topping` with fields called `pizza` and `name`. The `pizza` field should be a foreign key to `Pizza`, and `name` should be able to hold values such as `pineapple`, `Canadian bacon`, and `sausage`.

Register both models with the admin site, and use the site to enter some pizza names and toppings. Use the shell to explore the data you entered.

Making Pages: The Learning Log Home Page

Making web pages with Django consists of three stages: defining URLs, writing views, and writing templates. You can do these in any order, but in this project we'll always start by defining the URL pattern. A *URL pattern* describes the way the URL is laid out. It also tells Django what to look for when matching a browser request with a site URL, so it knows which page to return.

Each URL then maps to a particular view. The *view* function retrieves and processes the data needed for that page. The view function often renders the page using a *template*, which contains the overall structure of the page. To see how this works, let's make the home page for Learning Log. We'll define the URL for the home page, write its view function, and create a simple template.

Because we just want to ensure that Learning Log works as it's supposed to, we'll make a simple page for now. A functioning web app is fun to style when it's complete; an app that looks good but doesn't work well is pointless. For now, the home page will display only a title and a brief description.

Mapping a URL

Users request pages by entering URLs into a browser and clicking links, so we'll need to decide what URLs are needed. The home page URL is first: it's the base URL people use to access the project. At the moment, the base URL, `http://localhost:8000/`, returns the default Django site that lets us know the project was set up correctly. We'll change this by mapping the base URL to Learning Log's home page.